

Softmax + Cross-Entropy Gradient Derivation

Definitions

$$z_j = \sum_m W_{jm} x_m + b_j, \quad \hat{y}_i = \frac{e^{z_i}}{\sum_{k=1}^C e^{z_k}}$$
$$L = - \sum_{i=1}^C y_i \ln \hat{y}_i, \quad y \text{ one-hot}, \quad \sum_i y_i = 1$$

The summation indices (k, i, m) are dummy indices; j , the index we differentiate with respect to, is free.

Intermediate result

$$\ln \hat{y}_i = z_i - \ln \left(\sum_{k=1}^C e^{z_k} \right)$$
$$\frac{\partial}{\partial z_j} \ln \left(\sum_{k=1}^C e^{z_k} \right) = \frac{1}{\sum_{k=1}^C e^{z_k}} \frac{\partial}{\partial z_j} \left(\sum_{k=1}^C e^{z_k} \right) = \frac{e^{z_j}}{\sum_{k=1}^C e^{z_k}} = \hat{y}_j$$

Derivative of $\ln \hat{y}_i$ with respect to z_j

$$\frac{\partial \ln \hat{y}_i}{\partial z_j} = \frac{\partial z_i}{\partial z_j} - \frac{\partial}{\partial z_j} \ln \left(\sum_{k=1}^C e^{z_k} \right) = \delta_{ij} - \hat{y}_j$$

This splits into two cases:

$$\frac{\partial \ln \hat{y}_i}{\partial z_j} = \begin{cases} 1 - \hat{y}_j & (i = j) \\ -\hat{y}_j & (i \neq j) \end{cases}$$

Applying it to the loss

$$\begin{aligned} \frac{\partial L}{\partial z_j} &= - \sum_i y_i \frac{\partial \ln \hat{y}_i}{\partial z_j} = - \sum_i y_i (\delta_{ij} - \hat{y}_j) \\ &= - \sum_i y_i \delta_{ij} + \hat{y}_j \sum_i y_i = -y_j + \hat{y}_j \quad (\sum_i y_i = 1) \end{aligned}$$

$$\boxed{\frac{\partial L}{\partial z_j} = \hat{y}_j - y_j}$$

Gradient with respect to the parameters

$$\begin{aligned}\frac{\partial z_k}{\partial W_{jm}} &= \delta_{kj} x_m, & \frac{\partial z_k}{\partial b_j} &= \delta_{kj} \\ \frac{\partial L}{\partial W_{jm}} &= \sum_k \frac{\partial L}{\partial z_k} \frac{\partial z_k}{\partial W_{jm}} = (\hat{y}_j - y_j) x_m \\ \frac{\partial L}{\partial b_j} &= \hat{y}_j - y_j\end{aligned}$$

Batch of N examples

$$\begin{aligned}L &= \frac{1}{N} \sum_{n=1}^N L^{(n)} \\ \frac{\partial L}{\partial W_{jm}} &= \frac{1}{N} \sum_{n=1}^N \left(\hat{y}_j^{(n)} - y_j^{(n)} \right) x_m^{(n)}, & \frac{\partial L}{\partial b_j} &= \frac{1}{N} \sum_{n=1}^N \left(\hat{y}_j^{(n)} - y_j^{(n)} \right)\end{aligned}$$

In matrix form, for $X \in \mathbb{R}^{N \times M}$ and $\hat{Y}, Y \in \mathbb{R}^{N \times C}$:

$$\frac{\partial L}{\partial W} = \frac{1}{N} (\hat{Y} - Y)^\top X, \quad \frac{\partial L}{\partial b} = \frac{1}{N} (\hat{Y} - Y)^\top \mathbf{1}$$